MIL-R-24037/3(SHIPS)
29 May 1964
SUPERSEDING
MIL-R-19239(SHIPS)
16 January 1956

MILITARY SPECIFICATION

RADIACMETER - IM-107( )/PD

1. SCOPE

1.1 This specification covers an ion-chamber quartz-fiber dosimeter (radiacmeter), capable of measuring and indicating doses of gamma or X-radiation over the range 0 to 200 Roentgens (R).

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY
MIL-R-24037 - Radiacmeter, Quartz-fiber, Self-Indicating, General Specification.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The equipment shall be in accordance with MIL-R-24037 and as specified herein.

3.2 Scale.- The scale shall have 20 equal divisions, each division representing 10 Roentgens, and shall be marked with Arabic numerals. There shall be three major marking lines; 0, 100, and 200.

3.3 Range.- The range of the dosimeter shall be 0 to 200 Roentgens.

FSC 6665

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3.4 Leakage.- The electrical resistance between the charging electrode and the barrel shall be such that leakage in the absence of radiation shall be not more than 2.0 Roentgens per day for any relative humidity to 100 percent, and at any temperature between minus 65° and plus 131°F.

3.5 Leakage after heavy irradiation.- After a continuous or intermittent exposure totaling 5000 R given at dose rates up to 500,000 R per hour, electrical leakage in a 4-hour period starting not more than 10 minutes after exposure is completed, shall not exceed 5 percent of full scale. After a 48-hour period, the leakage requirement specified in 3.4 shall apply.

3.6 Accuracy.- The dosimetershall measure true gamma dose from a cobalt 60, radium or cesium 137 source within plus or minus 10 percent or plus or minus 5 Roentgens, whichever is larger. This accuracy requirement shall apply for any source which has been:

(a) Calibrated directly by the National Bureau of Standards.
(b) Calibrated by the contractor using Victoreen R meters, or equivalent approved by the bureau or agency concerned, which have in turn been calibrated by the National Bureau of Standards.
(c) Calibrated by a Government activity using Victoreen R meters, or equivalent approved by the bureau or agency concerned, which have been calibrated by the National Bureau of Standards.

Specifically included in this spread are the usual imprecisions involved in correlating source strengths with the standard National Bureau of Standards sources.

3.7 Energy dependence.- Except at the energies represented by cesium 137 and cobalt 60, where the requirements of 3.6 shall apply, the dosimeter shall indicate true gamma dose within plus or minus 15 percent or 5 R, whichever is larger, for X- or gamma radiation of any energy from 80 thousand electron volts (KEV) to the energy of cobalt 60 gammas. Over the range from 80 KEV to 50 KEV, the dosimeter response shall decrease smoothly; and at 50 KEV, the response shall be not greater than 50 percent of true dose, or 5 R, whichever is larger. As used here, 80 KEV and 50 KEV refer to effective energies of X-ray beams from a constant voltage and X-ray machine with appropriate tube voltage and filtration in accordance with normal National Bureau of Standards procedures.

3.8 Ion chamber saturation and rate dependence.- The radiacmeters, when subjected to dose rates up to 600,000 R per hour shall not vary from their normal calibration accuracy by more than plus or minus 2.0 R.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be as specified in MIL-R-24037, and as specified herein.

4.2 Leakage.- The radiacmeter shall be fully charged and tested for a period of 5 days at an ambient temperature of at least 100°F. The deflection of the electroscope fiber shall not exceed 10.0 R during this test. The radiacmeters may be stored under charge for 8 to 24 hours prior to this test if desired. The fiber shall be charged positively in this and all other tests.

4.3 Calibration at midpoint only.- A source, calibrated in accordance with 3.6, shall be used. Temperature may be ambient room temperature. A dose of 100 R shall be delivered to the radiacmeter. The reading shall be not greater than 110 R nor less than 90 R.

4.4 Calibration over the entire range.- The instrument shall be tested as specified in 4.3 at five points: 20 R, 60 R, 100 R, 140 R, and 180 R. Readings shall be as specified in 3.6.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-R-24037.

6. NOTES

6.1 The notes specified in MIL-R-24037 are applicable to this specification.

Preparing activity:
Navy - Ships
(Project 6665-N161Sh)

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